

D797

Artificial Intelligence and Machine Learning Foundations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A vendor pitches a deep learning product for reading invoices. The CFO, who has a law degree, asks how deep learning relates to AI and machine learning. Which answer is correct?

- A. Deep learning and machine learning are the same thing.
- B. Deep learning is a type of machine learning, which is a type of AI.
- C. They are three unrelated fields.
- D. AI is a type of machine learning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A chess program plays with hand-written rules and search and does not learn from data. Which label fits?

- A. Deep learning.
- B. Machine learning.
- C. Not AI at all, since all AI must learn from data.
- D. AI, but not machine learning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In the 1980s a bank built an expert system with 2,000 if-then rules written by loan officers. It handled common cases but was costly to update and failed on unusual ones. It ran on a mainframe in the basement. What limitation does this show?

- A. Hand-written rules are brittle and hard to scale; ML instead learns patterns from data.
- B. ML fixes this by needing no data at all.
- C. The mainframe was too slow.
- D. It needed 20,000 rules instead of 2,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A startup's image model performs millions of matrix multiplications per training step, and training on its CPU server takes three weeks. The office recently moved to a larger building. Which hardware change speeds training most, and why?

- A. More office space for servers.
- B. A CPU with a faster single core.
- C. GPUs, because they run thousands of arithmetic operations in parallel.
- D. Larger hard drives.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A warehouse robot learns to pick items by trying actions and getting a reward each time a pick succeeds. It was built by a team of five. Which type of learning is this?

- A. Team learning, since five people built it.
- B. Unsupervised learning.
- C. Supervised learning.
- D. Reinforcement learning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A retailer has customer purchase histories but no labels, and wants to discover natural customer groups. Which type of learning fits?

- A. None; nothing can be learned without labels.
- B. Supervised classification.
- C. Unsupervised learning, such as clustering.
- D. Reinforcement learning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A dataset of 10,000 patients has 30% of its income values missing. The data was exported on a Tuesday. What is the key risk of simply dropping every row with a missing income?

- A. None; dropping rows is always the safest choice.
- B. Export the data on a different day.
- C. Losing 3,000 rows and biasing the sample if the missing values are not random.
- D. Fill the gaps with zero instead.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

An age column has the values 20, 30, 40, and one missing value. Using mean imputation, what value fills the gap?

- A. 0
- B. 30
- C. 22.5
- D. 90

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

House sizes in a dataset range from 500 to 3,500 square feet. A 2,000-square-foot house was listed by a top agency. What is its min-max scaled value on a 0-1 scale?

- A. 0.5
- B. About 0.571
- C. It depends on the listing agency.
- D. About 0.667

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A feature has a minimum of 10 and a maximum of 50. What raw value has a min-max scaled value of 0.25?

- A. 12.5
- B. 20
- C. 160
- D. 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A student trains an image model on all 5,000 labeled photos and reports 99% accuracy on those same photos. The photos were taken with a new camera. What is wrong?

- A. It was tested on its own training data; accuracy must be measured on held-out data.
- B. Train longer to reach 100%.
- C. Nothing; 99% shows the model is excellent.
- D. Retake the photos with a better camera.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A company trains a hiring model on 10 years of its own hiring decisions, a period when most engineers hired were men. Training ran in a data center powered by renewable energy. What is the main ethical risk?

- A. The data center’s energy use.
- B. None; algorithms are objective.
- C. The model will favor women to balance history.
- D. The model learns and repeats the past bias against women.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A vendor pitches a deep learning product for reading invoices. The CFO, ~~who has a law degree,~~ asks how deep learning relates to AI and machine learning. Which answer is correct?

Target word: relationship of the terms **Decides:** Nested subsets.

A	Confusion set	Deep learning is one family of ML methods.
B	Correct	Each is a subset of the one before.
C	Adjacent category	They are nested, not separate.
D	Inverted operation	Reverses the nesting.

2. Answer D Concept 1 · transfer

A chess program plays with hand-written rules and search and does not learn from data. Which label fits?

Target word: classify a non-learning system **Decides:** AI without learning from data.

A	Adjacent category	No neural network is involved.
B	Confusion set	It learns nothing from data.
C	Absolute language	Much of AI history is rule-based.
D	Correct	Rule-based search is classic AI without learning.

3. Answer A Concept 2

In the 1980s a bank built an expert system with 2,000 if-then rules written by loan officers. It handled common cases but was costly to update and failed on unusual ones. ~~It ran on a mainframe in the basement.~~ What limitation does this show?

Target word: limitation of symbolic AI **Decides:** Hand-coded rules break outside what was written.

A	Correct	Rule systems cannot cover every case and need manual upkeep.
B	Inverted operation	ML depends on data.
C	Camouflage	Hardware speed was not the limitation.
D	Adjacent category	More rules deepen the upkeep problem.

4. Answer C Concept 3

A startup's image model performs millions of matrix multiplications per training step, and training on its CPU server takes three weeks. ~~The office recently moved to a larger building.~~ Which hardware change speeds training most, and why?

Target word: hardware choice **Decides:** Work is massively parallel matrix math.

A	Camouflage	Space does not change compute speed.
B	Confusion set	Helps serial work, not massive parallel math.
C	Correct	Matrix math spreads across many GPU cores.
D	Adjacent category	Storage size does not speed computation.

5. Answer D Concept 4

A warehouse robot learns to pick items by trying actions and getting a reward each time a pick succeeds. ~~It was built by a team of five.~~ Which type of learning is this?

Target word: learning type **Decides:** Trial and reward.

A	Camouflage	Team size is not a learning type.
B	Adjacent category	There is feedback, in the form of rewards.
C	Confusion set	No labeled correct answers are given.
D	Correct	Learns from rewards for actions.

6. Answer C Concept 4 • transfer

A retailer has customer purchase histories but no labels, and wants to discover natural customer groups. Which type of learning fits?

Target word: choose type from the data **Decides:** No labels; find groups.

A	Absolute language	Unsupervised methods work without labels.
B	Confusion set	Needs labeled groups, which do not exist.
C	Correct	Finds structure without labels.
D	Adjacent category	No actions or rewards are involved.

7. Answer C Concept 5

A dataset of 10,000 patients has 30% of its income values missing. ~~The data was exported on a Tuesday.~~ What is the key risk of simply dropping every row with a missing income?

Target word: risk of dropping rows **Decides:** 30% of rows would be lost.

A	Absolute language	It can shrink and skew the data.
B	Camouflage	The export day does not affect missing values.
C	Correct	Who is missing may differ from who is not.
D	Adjacent category	Zero is a false income and distorts the field.

8. Answer B Concept 5 • transfer

An age column has the values 20, 30, 40, and one missing value. Using mean imputation, what value fills the gap?

Target word: imputed value **Decides:** Mean of the known values.

A	Confusion set	Zero-filling is not mean imputation.
B	Correct	$(20 + 30 + 40) / 3 = 30$.
C	Wrong stem number	Divides by 4, counting the missing value.
D	Dropped element	The sum; drops the division.

9. Answer A Concept 6

House sizes in a dataset range from 500 to 3,500 square feet. A 2,000-square-foot house ~~was listed by a top agency.~~ What is its min-max scaled value on a 0-1 scale?

Target word: scaled value **Decides:** $(2,000 - 500) / 3,000$.

A	Correct	$1,500 / 3,000 = 0.5$.
B	Dropped element	$2,000 / 3,500$ drops the minimum.
C	Camouflage	The agency does not affect scaling.
D	Confusion set	$2,000 / 3,000$ divides by the range without subtracting the minimum.

10. Answer B Concept 6 · transfer

A feature has a minimum of 10 and a maximum of 50. What raw value has a min-max scaled value of 0.25?

Target word: reverse the scaling **Decides:** $x = \min + 0.25 \times (\max - \min)$.

A	Dropped element	0.25×50 drops the minimum.
B	Correct	$10 + 0.25 \times 40 = 20$.
C	Inverted operation	Divides 40 by 0.25 instead of multiplying.
D	Wrong stem number	The minimum itself.

11. Answer A Concept 7

A student trains an image model on all 5,000 labeled photos and reports 99% accuracy on those same photos. ~~The photos were taken with a new camera.~~ What is wrong?

Target word: evaluation flaw **Decides:** Tested on the training data.

A	Correct	Training accuracy says little about new data.
B	Inverted operation	Deepens memorization instead of checking generalization.
C	Dropped element	Ignores that the model has seen every test photo.
D	Camouflage	The camera is not the evaluation problem.

12. Answer D Concept 8

A company trains a hiring model on 10 years of its own hiring decisions, a period when most engineers hired were men. ~~Training ran in a data center powered by renewable energy.~~ What is the main ethical risk?

Target word: ethical risk **Decides:** Labels reflect past human choices.

A	Camouflage	A real issue elsewhere, but not the hiring risk here.
B	Absolute language	Models reflect the data they learn from.
C	Inverted operation	Models copy patterns; they do not correct them unaided.
D	Correct	Historical decisions become the pattern it copies.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	AI, ML, and deep learning	Deep learning is a subset of machine learning, which is a subset of AI; AI also includes systems that do not learn from data.	Reversing the nesting or treating the terms as synonyms.	How the three terms relate, or whether a rule-based system counts as AI.
2	Symbolic AI vs. machine learning	Symbolic (rule-based) AI encodes expert knowledge by hand and is brittle outside its rules; ML learns patterns from data.	Fixing brittleness by adding ever more rules.	An expert system that fails on unusual cases.
3	Hardware behind deep learning	GPUs run thousands of parallel arithmetic operations, which suits the matrix math of neural networks.	Crediting faster single CPU cores or storage.	Slow training dominated by matrix operations.
4	Supervised, unsupervised, reinforcement	Supervised learns from labeled examples; unsupervised finds structure without labels; reinforcement learns from rewards for actions.	Mixing up rewards with labels.	Labels, no labels, or trial and reward.
5	Handling missing data	Dropping incomplete rows loses data and can bias the sample; imputation fills gaps using values such as the mean of known values.	Always dropping, or filling with zero.	A share of missing values in a field.
6	Min-max scaling	Scaled value = $(x - \min) / (\max - \min)$, mapping the range to 0–1.	Dividing by max without subtracting the min.	A range and a raw value to normalize.
7	Evaluating on unseen data	A model must be tested on held-out data it never trained on to measure how it will generalize.	Reporting accuracy on the training data.	Accuracy measured on the same data used to train.
8	Bias from historical data	Models trained on past decisions learn and can repeat the biases in those decisions.	Believing algorithms are neutral by nature.	Training labels that come from past human choices.

Course scope (public description): Artificial Intelligence and Machine Learning Foundations explores foundational concepts and practical applications of artificial intelligence (AI) and machine learning (ML). It provides students with an understanding of the historical context and evolution of AI and the subsequent growth phases of AI technologies. This course will investigate the computational theories and logical frameworks that have shaped the AI landscape, examining how advancements in hardware have enabled the leap from basic ML to the complexities of deep learning. The course emphasizes the practicalities of AI and ML and gives insights into the critical role of data integrity and the techniques for data wrangling to feed into ML algorithms effectively. In addition to the technical skills, the course incorporates the societal implications of AI and ML, discussing current ethical considerations. This course aims to create a nuanced understanding of AI and ML, preparing students to thoughtfully consider the broader impacts of these technologies. There are no prerequisites for this course.